Fiscal Year:	FY 2025	**Task Last Updated:**	FY 01/05/2025
PI Name:	Yagoobi, Jamal Ph.D.		
Project Title:	Electrically Driven Liquid Film Flow Boiling: A Two-Phase Heat Transport Device Driven By Electric Conduction Mechanism		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE PI formerly at Illinois Institute of Technology; moved to Worcester Polytechnic Institute in January 2012. Info per PI (Ed., 1/17/2012)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	03/01/2022	**End Date:**	02/28/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	Not applicable		
COI Name (Institution):			
Grant/Contract No.:	80NSSC22K0676		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Task Description updated from original proposal (Ed., 7/18/23) The basic science and experimental effort covered by NASA Grant/Cooperative Agreement Number: NNX16AT09G entitled, “Electrically Driven Liquid Film Boiling - A Two-Phase Heat Transport Device Driven by Electric Conduction Mechanism” is essentially continuing with this new proposal. The specific tasks of the proposed project are: Task 1: Support the final design and analysis of the experiment conduction disc and heater, as well as detailed thermal analysis of the thermodynamics governing the experiment chamber. Develop and conduct ground-based experiment to determine the final flight matrix for potential study on the sub-orbital flights or International Space Station (ISS).		

Task Description:	Task 2: Complete tests as stated in the Science Requirements Document (SRD) (period: October 2021 to December 2022), with 2 mm liquid film thickness. The same will be conducted with the liquid film thickness of 3 mm. Upon completion of this effort, the combined results of 2 mm and 3 mm tests will used to finalize the test matrix to be conducted potentially aboard the sub-orbital flights or the ISS. Task 3: Continue with investigating the impact of the presence of non-condensable gas. The dielectrophoretic (DEP) force extracts the vapor bubble away from the heat surface. This is because of the difference between the vapor phase and liquid phase electric permittivity values. The vapor with a lower electric permittivity is attracted toward a lower electric field (i.e., DEP electrode) and the liquid with a higher electric permittivity is pushed toward the higher electric field (i.e., the heater). In the presence of a non-condensable gas (i.e., air), the DEP will extract the air away from the heater surface, thus minimizing or removing the impact of non-condensable gas on the liquid/vapor phase change taking place at the heater surface. A series of tests were conducted at Worcester Polytechnic Institute (WPI) by introducing air at different levels in 2023 and 2024. More detailed fundamental (theoretical and numerical) and applied work (experimental) in this area will be carried out in 2025 and 2026. In the experiments, the corresponding Psat and Tsat are measured to determine the amount of air introduced into the test vessel. In conclusion, the DEP force not only enhances the boiling and makes it feasible in the absence of gravity; it is also expected to eliminate the negative impact of the presence of non-condensable gas in the working fluid. Task 4: Numerical modelling of electrically driven liquid film flow in the presence of phase change in the radial configuration. The electrically driven liquid film flow generated by the electrohydrodynamic (EHD) conduction mechanism will be mathematically modeled to provide scientific results for the corresponding flow field and electric field. The problem in hand is very unique and challenging since the liquid film thickness changes in the direction of the flow due to condensation taking place at its liquid/vapor interface. Furthermore, the downstream boundary condition will need to take into account the presence of boiling and the DEP vapor extraction force. The numerical results will be compared in the presence and absence of gravity. Task 5: Detailed flow field measurements of electrically driven liquid film flow. The electrically driven liquid film flow field will be measured with the particle imaging velocimetry (PIV) unit. This will provide valuable data which will be used to compare them to the numerical predictions stated in Task 3. Initial PIV experiments will be conducted using our linear two-phase flow setup. Afterwards, modifications will be made to the radial unit, followed by corresponding flow field measurements with the PIV unit. Task 6: A detailed long-term study will be conducted to evaluate the system performance with respect to time and make sure that the effects of EHD conduction pumping of liquid film and the extraction of vapor bubbles do not degrade over time. Tentatively, a period of a full year will be considered for this continuous operation study. Task 7. The impact of oscillating the EHD conduction pumping on the heat transport capacity of the electrically driven liquid film flow boiling will be investigated. It is expected that there will be a critical frequency which will result in optimal operating conditions. This oscillation frequency will be related to the electric charge relaxation time. Task 8. Determine the impact of presence of electrical field on the performance of electronics during their cooling that is enhanced with externally imposed electrical field. This is essential in order to understand if the performance of the CPUs/chips will be impacted by the externally imposed electric field which is necessary for the operation of the EHD pumping and the activation of dielectrophoretic mechanism for extraction of vapor bubbles during boiling.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The rapid development of modern electronics has resulted in significant computational power in increasingly small packaging. However, a significant challenge remains in effectively controlling the temperature of electronic components, especially in harsh environments including space. With the revitalized interest in space development through NASA's Artemis Program and the increasing commercialization of space, development of improved thermal management technologies is needed with the objectives of reducing the mass and increasing the efficiency of heat transfer systems. Electrohydrodynamic (EHD) pumps have been identified as an advanced technology to support NASA's future human, robotic, and deep space missions. Electrohydrodynamic conduction pumps in particular have the advantage of being lightweight, with no moving parts, and are highly effective for thermal enhancement. From the progress made in this project, the body of knowledge and understanding of EHD conduction pumping applied for thermal management of electronics on earth as well as in space has been expanded. Though this research is targeted for electronics cooling in space, utilization of EHD conduction pumping for enhancement of existing terrestrial systems, such as vapor-compression cycle refrigeration, has the potential for improving the overall system efficiency. There is also the potential to improve the thermal performance of electronics, such as those in data centers found throughout the world. Thus, the future implementation of EHD has the potential for significant cost savings, and reduction in carbon footprint.
Task Progress:	This project entitled, "Electrically Driven Liquid Film Flow Boiling - A Two-phase Heat Transport Device Driven by Electric Conduction Mechanism", is being investigated to complete all the science tasks and to develop the hardware for potential sub-orbital flights and/or International Space Station, to conduct experiments on board to evaluate and validate the experimental data in the presence and absence of gravity. This progress report corresponds to the period of March 1, 2024, through February 28, 2025. However, the progress for the period of January 1, 2025 to February 28, 2025 is projected based on the expected tasks to be completed. The following tasks have successfully started in 2024 and will be continued in 2025. They are being conducted in the Multi-Scale Heat Transfer (MHT) Laboratory located at Worcester Polytechnic Institute (WPI). a) Comparison of Radial Flexible Electrohydrodynamic (EHD) Conduction Pump Critical Heat Flux (CHF) Evaluation to Silicon Wafer EHD Conduction Pump. b) Long-Term Testing of Electrically Driven Liquid Film Flow Boiling. c) Raspberry Pi CPU Performance in the Presence of Dielectrophoresis (DEP) Vapor Extraction Mechanism. d) Particle Image Velocimetry for Electrically Driven Liquid Film Flow Boiling.

	e) Single Phase Numerical Evaluation of Heat Transfer Performance of Oscillating EHD Conduction Pumping.
Bibliography Type:	Description: (Last Updated: 01/30/2026)
Abstracts for Journals and Proceedings	Castaneda AJ, Yagoobi J. "Heat transfer characteristics of oscillating electrohydrodynamic liquid flow." SHTC 2024, ASME Summer Heat Transfer Conference, Anaheim, California, July 15-17, 2024. SHTC 2024 Final Program, American Society of Mechanical Engineers. 2024. Paper No: HT2024-131411. https://doi.org/10.1115/HT2024-131411 , Jul-2024
Abstracts for Journals and Proceedings	Podlaski L, Castaneda A, Yagoobi J. "Flexible radial electrohydrodynamic conduction pump liquid film flow boiling for long-term performance testing." 2024 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 9-13, 2024. Abstracts. 2024 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 9-13, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Castaneda A, Yagoobi J. "Oscillating electrohydrodynamic conduction pumping in liquid – A numerical study." 2024 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 9-13, 2024. Abstracts. 2024 Annual Meeting of the Electrostatics Society of America, Ottawa, Canada, June 9-13, 2024. , Jun-2024